# Calculated Variables
in the
# 2025 Data File
of the
# Behavioral Risk Factor Surveillance System

***(Version #5 - Revised: June 26, 2026)***






# Introduction

This document provides information on calculated variables for the 2025 Behavioral Risk Factor Surveillance System survey. These variables are calculated from responses to questions in the survey. There are three types of calculated variables:

1. Variables used to stratify and weight the data (not included in this document).

2. Intermediate variables, which are derived from a question response and are used to calculate some other variable or risk factor. Example: WTKG3 is derived from the WEIGHT2 variable in the survey. WTKG3 is then used to calculate the body mass index variable (_BMI5). Most—but not all—of the intermediate variables end with an underscore such as FTJUDA2_.

3. Variables used to categorize or classify respondents. Most of these begin with an underscore such as _BMI5. Exceptions are: _DENSTR2, _GEOSTR, and _STATE, which are determined before the interview. Some of the calculated variables group continuous variables—such as weight, age, or body mass index—into categories. Other calculated variables regroup non-continuous variables to simplify analyses. The common focus of these variables is on health behaviors that are associated with a risk of illness or injury.

The tables in this report include a description of what the responses mean and a copy of the code used to calculate these variables in SAS®. The syntax of the code, as given, may or may not work in the particular statistical program that you are using. Not all variables used to derive the calculated variables are in the public access dataset.

| **Section 1: Health Status** | | |
|---|---|---|
| **_RFHLTH** | ***Calculated variable for adults with good or better health.* _RFHLTH is derived from GENHLTH.** | |
| 1 | Good or Better Health | Respondents who reported having excellent, very good, or good health. (GENHLTH =1, 2, 3) |
| 2 | Fair or Poor Health | Respondents who reported having fair or poor health. (GENHLTH =4, 5) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn't know, refused to answer, or had missing responses for the general health status question. (GENHLTH =7, 9, missing) |
| | **SAS Code:** | `IF 4 LE GENHLTH LE 5 THEN _RFHLTH=2;`<br>` ELSE IF 1 LE GENHLTH LE 3 THEN _RFHLTH=1;`<br>` ELSE _RFHLTH=9;` |

| **Section 2: Healthy Days** | | |
|---|---|---|
| **_PHYS14D** | ***Calculated variable for 3 levels of not-good physical health status: 0 days, 1-13 days, 14-30 days.* _PHYS14D is derived from PHYSHLTH.** | |
| 1 | Zero days when physical health not good | Respondents who reported no days when their physical health was not good (PHYSHLTH=88) |
| 2 | 1-13 days when physical health not good | Respondents who reported 1-13 days when their physical health was not good (1 </= PHYSHLTH </= 13) |
| 3 | 14+ days when physical health not good | Respondents who reported 14 or more days when their physical health was not good (14 </= PHYSHLTH </= 30) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they didn´t know, refused, or had missing values for PHYSHLTH (PHYSHLTH=77,99, or missing) |
| | **SAS Code:** | `IF PHYSHLTH IN (77,99,.) THEN _PHYS14D=9;`<br>`ELSE IF PHYSHLTH=88 THEN _PHYS14D=1;`<br>`ELSE IF 1 LE PHYSHLTH LE 13 THEN _PHYS14D=2;`<br>`ELSE _PHYS14D=3;` |

| **Section 2: Healthy Days** | | |
|---|---|---|
| **_MENT14D** | ***Calculated variable for 3 levels of not-good mental health status: 0 days, 1-13 days, 14-30 days.*** **_MENT14D is derived from MENTHLTH.** | |
| 1 | Zero days when mental health not good | Respondents who reported no days when their mental health was not good (MENTHLTH=88) |
| 2 | 1-13 days when mental health not good | Respondents who reported 1-13 days when their mental health was not good (1 </= MENTHLTH </= 13) |
| 3 | 14+ days when mental health not good | Respondents who reported 14 or more days when their mental health was not good (14 </= MENTHLTH </= 30) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they didn't know, refused, or had missing values for MENTHLTH (MENTHLTH=77,99, or missing) |
| | **SAS Code:** | `IF MENTHLTH IN (77,99,.) THEN _MENT14D=9;`<br>`ELSE IF MENTHLTH=88 THEN _MENT14D=1;`<br>`ELSE IF 1 LE MENTHLTH LE 13 THEN _MENT14D=2;`<br>`ELSE _MENT14D=3;` |

| **Section 3: Health Care Access** | | |
|---|---|---|
| **_HLTHPL2** | ***Calculated variable for adults who had some form of health insurance.*** **_HLTHPL2 is derived from PRIMINS2.** | |
| 1 | Have some form of insurance | Respondents who said they had some form of health insurance (PRIMINS2=1, 2, 3, 4, 5, 6, 7, 8, 9, 10) |
| 2 | Do not have some form of health insurance | Respondents who said they did not have some form of health insurance (PRIMINS2=88) |
| 9 | Don´t know, refused, or missing insurance response | Respondents who refused, didn´t know, or were missing a response to having some form of health insurance (PRIMINS2=77, 99 or missing) |
| | **SAS Code:** | `IF PRIMINS2 in (1,2,3,4,5,6,7,8,9,10) THEN _HLTHPL2=1;`<br>`ELSE IF PRIMINS2=88 THEN _HLTHPL2=2;`<br>`ELSE _HLTHPL2=9;` |

| **Section 3: Health Care Access** | | |
|---|---|---|
| **_HCVU654** | ***Calculated variable for respondents aged 18-64 who have any form of health insurance.* _HCVU654 is derived from AGE and PRIMINS2.** | |
| 1 | Have some form of health insurance | Respondents who reported having some form of health insurance (18 </= AGE </= 64 and PRIMINS2=1, 2, 3, 4, 5, 6, 7, 8, 9, 10) |
| 2 | Do not have any form of health insurance | Respondents who reported not having any form of health insurance (18 </= AGE </= 64 and PRIMINS2=88) |
| 9 | Don't know/ Not Sure, Refused or Missing | Respondents who reported that they didn´t know, were not sure, refused to report, or had missing responses for having health care coverage (18 </= AGE </= 64 and PRIMINS2=77, 99, or missing or AGE = > 65) |
| | **SAS Code:** | `IF 18 LE AGE LE 64 THEN DO;`<br>`IF PRIMINS2 in (1,2,3,4,5,6,7,8,9,10) THEN _HCVU654=1;`<br>`ELSE IF PRIMINS2=88 THEN _HCVU654=2;`<br>`ELSE _HCVU654=9;`<br>`END;`<br>`ELSE _HCVU654 = 9;` |

| **Section 4: Exercise (Physical Activity)** | | |
|---|---|---|
| **_TOTINDA** | ***Calculated variable for adults who reported doing physical activity or exercise during the past 30 days other than their regular job.* _TOTINDA is derived from EXERANY2.** | |
| 1 | Had physical activity or exercise | Respondents who reported doing any physical activity or exercise. (EXERANY2=1) |
| 2 | No physical activity or exercise in last 30 days | Respondents who reported doing no physical activity or exercise. (EXERANY2=2) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they didn't know or refused to answer, and those with missing responses for the physical activity/exercise question. (EXERANY2=7, 9, missing) |
| | **SAS Code:** | `IF EXERANY2 IN (1) THEN _TOTINDA=1;`<br>`ELSE IF EXERANY2 IN (2) THEN _TOTINDA=2;`<br>`ELSE IF EXERANY2 IN (.,7,9) THEN _TOTINDA=9;` |

| **Section 5: Hypertension Awareness** | | |
|---|---|---|
| **_RFHYPE6** | ***Calculated variable for adults who have been told they have high blood pressure by a doctor, nurse, or other health professional.* _RFHYPE6 is derived from BPHIGH6.** | |
| 1 | No | Respondents who were not told their pressure is high by a health professional (BPHIGH6=2, 3, or 4) |
| 2 | Yes | Respondents who were told their pressure is high by a health professional (BPHIGH6=1) |
| 9 | Don't know/ Not Sure/ Refused/ Missing | Respondents who reported they didn´t know if they were told if their blood pressure is high, those who refused to answer if they were told if their blood pressure is high, and those with missing responses (BPHIGH6=7, 9, or missing) |
| | **SAS Code:** | `IF BPHIGH6 = 1 THEN _RFHYPE6=2;`<br>`ELSE IF BPHIGH6 = 2 THEN _RFHYPE6=1;`<br>`ELSE IF BPHIGH6 = 3 THEN _RFHYPE6=1;`<br>`ELSE IF BPHIGH6 = 4 THEN _RFHYPE6=1;`<br>`ELSE IF BPHIGH6 IN (.,7,9) THEN _RFHYPE6=9 ;` |

| **Section 6: Cholesterol Awareness** | | |
|---|---|---|
| **_CHOLCH3** | ***Calculated variable for cholesterol check within past five years.* _CHOLCH3 is derived from CHOLCHK3.** | |
| 1 | Had cholesterol checked in past 5 years | Respondents who reported having had their cholesterol checked within the past five years (CHOLCH3=2, 3, 4, 5, or 6) |
| 2 | Did not have cholesterol checked in past 5 years | Respondents who reported not having had their cholesterol checked within the past five years (CHOLCH3=8) |
| 3 | Have never had cholesterol checked | Respondents who reported never having had their cholesterol checked (CHOLCH3=1) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know if they had their cholesterol checked by a health professional, those who refused to answer if they had their cholesterol checked by a health professional, and those with missing responses (CHOLCH3=7, 9 or missing) |
| | **SAS Code:** | `IF CHOLCHK3=1 THEN _CHOLCH3=3;`<br>`ELSE IF CHOLCHK3 in (2, 3, 4, 5, 6) THEN _CHOLCH3=1;`<br>`ELSE IF CHOLCHK3 = 8 THEN _CHOLCH3=2;`<br>`ELSE _CHOLCH3=9;` |

| **Section 6: Cholesterol Awareness** | | |
|---|---|---|
| **_RFCHOL3** | ***Calculated variable for adults who have had their cholesterol checked and have been told by a doctor, nurse, or other health professional that it was high.*** **_RFCHOL3 is derived from CHOLCHK3 and TOLDHI3.** | |
| 1 | No | Respondents who reported having had their blood cholesterol checked but had not been told it was high (CHOLCHK3=2,3,4,5,6 or 8 and TOLDHI3=2) |
| 2 | Yes | Respondents who reported having had their blood cholesterol checked and had been told that they have high blood cholesterol (CHOLCHK3=2,3,4,5,6 or 8 and TOLDHI3=1) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know if they had their blood cholesterol checked, those who reported they didn´t know if they have been told their blood cholesterol was high, those who refused to answer if they had their blood cholesterol checked, those who refused to answer if they had been told that their blood cholesterol was high, and those with missing responses (CHOLCHK3=2,3,4,5,6 or 8 and TOLDHI3=7,9,or missing) |
| . | Missing | Respondents who reported they have not had their blood cholesterol checked (CHOLCHK3=1,7,9 or missing) |
| | **SAS Code:** | `IF CHOLCHK3 in (2,3,4,5,6,8) AND TOLDHI3=1 THEN _RFCHOL3=2;`<br>`ELSE IF CHOLCHK3 in (2,3,4,5,6,8) AND TOLDHI3=2 THEN _RFCHOL3=1;`<br>`ELSE IF CHOLCHK3 in (2,3,4,5,6,8) AND TOLDHI3 in (7,9,.) THEN _RFCHOL3=9;` |

| **Section 7: Chronic Health Conditions** | | |
|---|---|---|
| **_MICHD** | ***Calculated variable for respondents who have ever reported having coronary heart disease (CHD) or myocardial infarction (MI).*** **_MICHD is derived from CVDINFR4, and CVDCRHD4.** | |
| 1 | Reported having MI or CHD | Respondents who reported having had MI or CHD (CVDINFR4=1 OR CVDCRHD4=1) |
| 2 | Did not report having MI or CHD | Respondents who reported not having had MI and CHD (CVDINFR4=2 AND CVDCRHD4=2) |
| . | Not asked or Missing | Respondents who reported they didn´t know, refused or had a missing value for the MI or CHD questions (CVDINFR4=7, 9 OR MISSING OR CVDCRHD4=7, 9, OR MISSING) |
| | **SAS Code:** | `IF CVDINFR4=1 OR CVDCRHD4=1 THEN _MICHD=1;`<br>`ELSE IF CVDINFR4=2 AND CVDCRHD4=2 THEN _MICHD=2;` |

| Section 7: Chronic Health Conditions | | |
|---|---|---|
| **_LTASTH1** | ***Calculated variable for adults who have ever been told they have asthma.*** **_LTASTH1 is derived from ASTHMA3.** | |
| 1 | No | Respondents who have not been told by a doctor, nurse, or health professional that they had asthma. (ASTHMA3=2) |
| 2 | Yes | Respondents who have been told by a doctor, nurse, or health professional that they had asthma. (ASTHMA3=1) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they did not know if they had been told by a doctor, nurse, or health professional that they had asthma, those who refused to answer if they had been told by a doctor, nurse, or health professional that they had asthma, or those with missing responses. (ASTHMA3=7, 9, missing) |
| | **SAS Code:** | `IF ASTHMA3=1 THEN _LTASTH1=2;`<br>`ELSE IF ASTHMA3=2 THEN _LTASTH1=1;`<br>`ELSE _LTASTH1=9;` |

| Section 7: Chronic Health Conditions | | |
|---|---|---|
| **_CASTHM1** | ***Calculated variable for adults who have been told they currently have asthma.*** **_CASTHM1 is derived from ASTHMA3 and ASTHNOW.** | |
| 1 | No | Respondents who have not been told by a doctor, nurse, or health professional that they had asthma or do not still have asthma. (ASTHMA3=2 or ASTHMA3=1 and ASTHNOW=2) |
| 2 | Yes | Respondents who have been told by a doctor, nurse, or health professional that they had asthma and that they still have asthma. (ASTHMA3=1 and ASTHNOW=1) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they did not know if they had been told by a doctor, nurse, or health professional that they had asthma, those who refused to answer if they had been told by a doctor, nurse, or health professional that they had asthma, those who did not know if they still had asthma, those who refused to answer if they still had asthma, or those with missing responses. (ASTHMA3=7, 9, missing; or ASTHNOW=7, 9, missing) |
| | **SAS Code:** | `IF ASTHMA3=2 THEN _CASTHM1=1;`<br>`ELSE IF ASTHMA3=1 AND ASTHNOW=1 THEN _CASTHM1=2;`<br>`ELSE IF ASTHMA3=1 AND ASTHNOW=2 THEN _CASTHM1=1;`<br>`ELSE _CASTHM1=9;` |

| Section 7: Chronic Health Conditions | | |
|---|---|---|
| **_ASTHMS1** | ***Calculated variable for computed asthma status.*** **_ASTHMS1 is derived from ASTHMA3 and ASTHNOW.** | |
| 1 | Current | Respondents who have been told by a doctor, nurse, or health professional that they had asthma and that they still have asthma. (ASTHMA3=1and ASTHNOW=1) |
| 2 | Former | Respondents who have been told by a doctor, nurse, or health professional that they had asthma but do not still have asthma. (ASTHMA3=1 and ASTHNOW=2) |
| 3 | Never | Respondents who have not been told by a doctor, nurse, or health professional that they had asthma. (ASTHMA3=2) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn't know if they had been told by a doctor, nurse, or health professional that they had asthma, those who refused to answer if they had been told by a doctor, nurse, or health professional that they had asthma, those who didn't know if they still had asthma, those who refused to answer if they still had asthma, or those with missing responses. (ASTHMA3=7, 9, missing; or ASTHNOW=7, 9, missing) |
| | **SAS Code:** | `IF ASTHMA3=1 AND ASTHNOW=1 THEN _ASTHMS1=1;`<br>`ELSE IF ASTHMA3=1 AND ASTHNOW=2 THEN _ASTHMS1=2;`<br>`ELSE IF ASTHMA3=2 THEN _ASTHMS1=3;`<br>`ELSE _ASTHMS1=9;` |

| Section 7: Chronic Health Conditions | | |
|---|---|---|
| **_DRDXAR2** | ***Calculated variable for respondents who have had a doctor diagnose them as having some form of arthritis.*** **_DRDXAR2 is derived from HAVARTH4.** | |
| 1 | Diagnosed with arthritis | Respondents who have been told by a doctor they had arthritis (HAVARTH4=1) |
| 2 | Not diagnosed with arthritis | Respondents who have not been told by a doctor they had arthritis (HAVARTH4=2) |
| . | Don´t know/ Not Sure/ Refused/ Missing | Respondents who reported they didn´t know if they had been told by a doctor they had arthritis, those who refused to answer if they had been told by a doctor they had arthritis, and those with missing responses (HAVARTH4=7,9, or missing) |
| | **SAS Code:** | `IF HAVARTH4 = 1 THEN _DRDXAR2=1;`<br>`ELSE IF HAVARTH4 = 2 THEN _DRDXAR2=2;`<br>`ELSE IF HAVARTH4 IN (7,9,.) THEN _DRDXAR2=.;` |

| Section 8: Demographics | | |
|---|---|---|
| **_SEX** | ***Calculated variable for calculated sex variable.*** **_SEX is derived from BIRTHSEX and SEXVAR** | |
| 1 | Male | Male respondent (BIRTHSEX=1 or BIRTHSEX notin (1,2) and SEXVAR=1) |
| 2 | Female | Female respondent (BIRTHSEX=2 or BIRTHSEX notin (1,2) and SEXVAR=2) |
| | **SAS Code:** | `IF BIRTHSEX IN (1,2) THEN DO;`<br>`_SEX=BIRTHSEX;`<br>`END;`<br>`ELSE DO;`<br>`_SEX=SEXVAR;`<br>`END;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_AGEG5YR** | ***Calculated variable for fourteen-level age category.*** **_AGEG5YR is derived from AGE.** | |
| 1 | Age 18 to 24 | Respondents with reported age between 18 and 24 years (18 </= AGE </= 24) |
| 2 | Age 25 to 29 | Respondents with reported age between 25 and 29 years (25 </= AGE </= 29) |
| 3 | Age 30 to 34 | Respondents with reported age between 30 and 34 years (30 </= AGE </= 34) |
| 4 | Age 35 to 39 | Respondents with reported age between 35 and 39 years (35 </= AGE </= 39) |
| 5 | Age 40 to 44 | Respondents with reported age between 40 and 44 years (40 </= AGE </= 44) |
| 6 | Age 45 to 49 | Respondents with reported age between 45 and 49 years (45 </= AGE </= 49) |
| 7 | Age 50 to 54 | Respondents with reported age between 50 and 54 years (50 </= AGE </= 54) |
| 8 | Age 55 to 59 | Respondents with reported age between 55 and 59 years (55 </= AGE </= 59) |
| 9 | Age 60 to 64 | Respondents with reported age between 60 and 64 years (60 </= AGE </= 64) |
| 10 | Age 65 to 69 | Respondents with reported age between 65 and 69 years (65 </= AGE </= 69) |
| 11 | Age 70 to 74 | Respondents with reported age between 70 and 74 years (70 </= AGE </= 74) |
| 12 | Age 75 to 79 | Respondents with reported age between 75 and 79 years (75 </= AGE </= 79) |
| 13 | Age 80 or older | Respondents with reported age between 80 and 99 years (80 </= AGE </= 99) |
| 14 | Don’t know/ Refused/ Missing | Respondents who reported they didn´t know, were not sure, refused to report or had missing responses for their age. (AGE=7, 9, missing) |
| | **SAS Code:** | |

```
IF 18 LE AGE LE 24 THEN _AGEG5YR = 1;
 ELSE IF 25 LE AGE LE 29 THEN _AGEG5YR = 2;
 ELSE IF 30 LE AGE LE 34 THEN _AGEG5YR = 3;
 ELSE IF 35 LE AGE LE 39 THEN _AGEG5YR = 4;
 ELSE IF 40 LE AGE LE 44 THEN _AGEG5YR = 5;
ELSE IF 45 LE AGE LE 49 THEN _AGEG5YR = 6;
 ELSE IF 50 LE AGE LE 54 THEN _AGEG5YR = 7;
 ELSE IF 55 LE AGE LE 59 THEN _AGEG5YR = 8;
 ELSE IF 60 LE AGE LE 64 THEN _AGEG5YR = 9;
 ELSE IF 65 LE AGE LE 69 THEN _AGEG5YR = 10;
ELSE IF 70 LE AGE LE 74 THEN _AGEG5YR = 11;
 ELSE IF 75 LE AGE LE 79 THEN _AGEG5YR = 12;
 ELSE IF 80 LE AGE LE 99 THEN _AGEG5YR = 13;
 ELSE _AGEG5YR = 14;
```

| **Section 8: Demographics** | | |
|---|---|---|
| **_AGE65YR** | ***Calculated variable for two-level age category.* _AGE65YR is derived from AGE.** | |
| 1 | Age 18 to 64 | Respondents with reported ages 18–64. (18 </= AGE </= 64) |
| 2 | Age 65 or older | Respondents with reported ages 65–99. (65 >/= AGE >/= 99) |
| 3 | Don't know/ Refused/ Missing | Respondents who reported they didn´t know, were not sure, refused, or had a missing value for AGE. (AGE=7,9,or missing) |
| | **SAS Code:** | `IF 18 LE AGE LE 64 THEN _AGE65YR=1;`<br>`ELSE IF 65 LE AGE LE 99 THEN _AGE65YR=2;`<br>`ELSE _AGE65YR = 3;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_AGE80** | ***Calculated variable for imputed age value collapsed above 80.* _AGE80 is derived from _IMPAGE.** | |
| 18 - 24 | Imputed Age 18 to 24 | Respondents with reported/Imputed Age between 18 and 24 years (18 </= Imputed Age </= 24) |
| 25 - 29 | Imputed Age 25 to 29 | Respondents with reported/Imputed Age between 25 and 29 years (25 </= Imputed Age </= 29) |
| 30 - 34 | Imputed Age 30 to 34 | Respondents with reported/Imputed Age between 30 and 34 years (30 </= Imputed Age </= 34) |
| 35 - 39 | Imputed Age 35 to 39 | Respondents with reported/Imputed Age between 35 and 39 years (35 </= Imputed Age </= 39) |
| 40 - 44 | Imputed Age 40 to 44 | Respondents with reported/Imputed Age between 40 and 44 years (40 </= Imputed Age </= 44) |
| 45 - 49 | Imputed Age 45 to 49 | Respondents with reported/Imputed Age between 45 and 49 years (45 </= Imputed Age </= 49) |
| 50 - 54 | Imputed Age 50 to 54 | Respondents with reported/Imputed Age between 50 and 54 years (50 </= Imputed Age </= 54) |
| 55 - 59 | Imputed Age 55 to 59 | Respondents with reported/Imputed Age between 55 and 59 years (55 </= Imputed Age </= 59) |
| 60 - 64 | Imputed Age 60 to 64 | Respondents with reported/Imputed Age between 60 and 64 years (60 </= Imputed Age </= 64) |
| 65 - 69 | Imputed Age 65 to 69 | Respondents with reported/Imputed Age between 65 and 69 years (65 </= Imputed Age </= 69) |
| 70 - 74 | Imputed Age 70 to 74 | Respondents with reported/Imputed Age between 70 and 74 years (70 </= Imputed Age </= 74) |
| 75 - 79 | Imputed Age 75 to 79 | Respondents with reported/Imputed Age between 75 and 79 years (75 </= Imputed Age </= 79) |
| 80 - 99 | Imputed Age 80 or older | Respondents with reported/Imputed Age between 80 and 99 years (80 </= Imputed Age </= 99) |
| | **SAS Code:** | `IF 18 LE _IMPAGE LE 80 THEN _AGE80=_IMPAGE;`<br>`ELSE IF _IMPAGE GE 80 THEN _AGE80=80;` |

_AGE80 may be best, input is just age rather than a group. Easier to interpret and can be continuous

| Section 8: Demographics | | |
|---|---|---|
| **_AGE_G** | ***Calculated variable for six-level imputed age category.* _AGE_G is derived from _IMPAGE (imputed age).** | |
| 1 | Age 18 to 24 | Respondents with imputed ages between 18–24 years of age. (18 </= _IMPAGE </= 24) |
| 2 | Age 25 to 34 | Respondents with imputed ages between 25–34 years of age. (25 </= _IMPAGE </= 34) |
| 3 | Age 35 to 44 | Respondents with imputed ages between 35–44 years of age. (35 </= _IMPAGE </= 44) |
| 4 | Age 45 to 54 | Respondents with imputed ages between 45–54 years of age. (45 </= _IMPAGE </= 54) |
| 5 | Age 55 to 64 | Respondents with imputed ages between 55–64 years of age. (55 </= _IMPAGE </= 64) |
| 6 | Age 65 or older | Respondents with imputed ages between 65–99 years of age. (_IMPAGE = > 65) |
| | **SAS Code:** | `IF (18<=_IMPAGE<=24) THEN _AGE_G = 1;`<br>`ELSE IF (25<=_IMPAGE<=34) THEN _AGE_G = 2;`<br>`ELSE IF (35<=_IMPAGE<=44) THEN _AGE_G = 3;`<br>`ELSE IF (45<=_IMPAGE<=54) THEN _AGE_G = 4;`<br>`ELSE IF (55<=_IMPAGE<=64) THEN _AGE_G = 5;`<br>`ELSE IF (_IMPAGE >= 65) THEN _AGE_G = 6;` |

| Section 8: Demographics | | |
|---|---|---|
| **HTIN4** | ***Calculated variable for reported height in inches.* HTIN4 is derived from HEIGHT3. HTIN4 is calculated by adding the foot portion of HEIGHT3 multiplied by 12, to the inch portion.** | |
| 36 - 95 | Height in inches | Respondents calculated height in inches. (HTIN4=(height in feet x 12) + height in inches) |
| . | Don't know/ Refused/ Not asked or Missing | Respondents who reported they didn´t know, were not sure, refused or had missing responses for their height. |
| | **SAS Code:** | `IF 300<=HEIGHT3<=311 THEN HTIN4=((HEIGHT3-300)+36);`<br>`ELSE IF 400<=HEIGHT3<=411 THEN HTIN4=((HEIGHT3-400)+48);`<br>`ELSE IF 500<=HEIGHT3<=511 THEN HTIN4=((HEIGHT3-500)+60);`<br>`ELSE IF 600<=HEIGHT3<=611 THEN HTIN4=((HEIGHT3-600)+72);`<br>`ELSE IF 700<=HEIGHT3<=711 THEN HTIN4=((HEIGHT3-700)+84);` |

| Section 8: Demographics | | |
|---|---|---|
| **HTM4** | ***Calculated variable for reported height in meters.* HTM4 is derived from the variable HTIN4 by multiplying HTIN4 by 2.54 cm per in and dividing by 100 cm per meter. HTM4 is derived from HEIGHT2 metric values by dividing by 100.** | |
| 91 - 244 | Height in meters [2 implied decimal places] | Respondents reported or calculated height in meters. (HTM4=HTIN4 x 0.0254 or HTM4 = (HEIGHT3 - 9000) ÷ 100) |
| . | Don't know/ Refused/ Not asked or Missing | Respondents who reported they didn´t know, were not sure, refused or had missing responses for their height. |
| | **SAS Code:** | `IF 300 <= HEIGHT3 <= 711 THEN HTM4=HTIN4*0.0254;`<br>`ELSE IF 9091 <= HEIGHT3 < 9244 THEN HTM4=(HEIGHT3-9000)/100;` |

Use height in meters - has half the missing values

Be sure to get weight from non-calculated variabels



| Section 8: Demographics | | |
|---|---|---|
| **WTKG3** | ***Calculated variable for reported weight in kilograms.* WTKG3 is derived from WEIGHT2 by multiplying WEIGHT2 by 0.4535924 kg per lb.** | |
| 2300 - 29500 | Weight in kilograms [2 implied decimal places] | Respondents reported or calculated weight in kilograms. |
| . | Don't know/ Refused/ Not asked or Missing | Respondents who reported they didn't know, were not sure, or refused or had missing responses for their weight. |
| | **SAS Code:** | `** CONVERSION FACTOR = 0.4535924 kg/lb **;`<br>` IF WEIGHT2 NOT IN (777,999,7777,9999,.) THEN DO;`<br>` IF 0050 LE WEIGHT2 < 0650 THEN WTKG3=WEIGHT2*0.4535924;`<br>` ELSE IF 9023 LE WEIGHT2 < 9295 THEN WTKG3=WEIGHT2-9000;`<br>`END;` |

| Section 8: Demographics | | |
|---|---|---|
| **_BMI5** | ***Calculated variable for body mass index (BMI).* _BMI5 is derived from WTKG3 and HTM4. It is calculated by dividing WTKG3 by HTM4².** | |
| 1 - 9999 | 1 or greater | Respondents calculated body mass index (BMI) {units=kilograms per meter squared}. (_BMI5 = WTKG3 / (HTM4xHTM4)) |
| . | Don't know/ Refused/ Missing | Respondents who had a missing value for their height in meters or weight in kilograms. (WTKG3=missing or HTM4=missing or _BMI5 < 12.00 or _BMI5 >/= 100 or PREGNANT=1) |
| | **SAS Code:** | `IF (WTKG3 NOTIN (.)) AND (HTM4 NOTIN (.)) THEN _BMI5=WTKG3/(HTM4 ** 2);`<br>` ELSE _BMI5=.;`<br>`IF _BMI5 NE . THEN _BMI5=ROUND(_BMI5,.01);`<br>` IF _BMI5 > 99.99 THEN _BMI5=.;`<br>` IF _BMI5 < 12.00 THEN _BMI5=.;`<br>`IF PREGNANT=1 THEN _BMI5=.;` |

| Section 8: Demographics | | |
|---|---|---|
| **_BMI5CAT** | ***Calculated variable for four categories of body mass index (BMI).* _BMI5CAT is derived from _BMI5.** | |
| 1 | Underweight | Respondents classified as underweight based on body mass index. (_BMI5 < 18.50) |
| 2 | Normal Weight | Respondents classified as normal weight based on body mass index. (18.50 </= _BMI5 < 25.00) |
| 3 | Overweight | Respondents classified as overweight based on body mass index. (25.00 </= _BMI5 < 30.00) |
| 4 | Obese | Respondents classified as obese based on body mass index. (30.00 </= _BMI5 < 99.99) |
| . | Don't know/ Refused/ Missing | Respondents with an unknown, refused, or missing value for body mass index. (_BMI5=.) |
| | **SAS Code:** | `IF (0.00 LE _BMI5 < 18.50) THEN _BMI5CAT=1;`<br>` ELSE IF (18.50 LE _BMI5 < 25.00) THEN _BMI5CAT=2;`<br>` ELSE IF (25.00 LE _BMI5 < 30.00) THEN _BMI5CAT=3;`<br>`ELSE IF _BMI5 GE 30.00 THEN _BMI5CAT=4;` |

Can choose cat/cont for BMI

| Section 8: Demographics | | |
|---|---|---|
| **_RFBMI5** | ***Calculated variable for adults who have a body mass index greater than 25.00 (overweight or obese). _RFBMI5 is derived from _BMI5.*** | |
| 1 | No | Respondents not classified as overweight or obese based on body mass index. (12 </= _BMI5 < 25.00) |
| 2 | Yes | Respondents classified as overweight or obese based on body mass index. (25.00 </= _BMI5 </= 99.99) |
| 9 | Don't know/ Refused/ Missing | Respondents with an unknown, refused, or missing value for body mass index. (_BMI5=missing) |
| | **SAS Code:** | `IF (12.00 LE _BMI5 < 25.00) THEN _RFBMI5=1;`<br>` ELSE IF (25.00 <= _BMI5 < 99.99) THEN _RFBMI5=2;`<br>` ELSE _RFBMI5=9;`<br>`** Round off HTM4, WTKG3 and _BMI5 to 2 decimal places and remove the decimal **;`<br>` HTIN4 = round(HTIN4,1);`<br>` HTM4 = round((HTM4*100),1);`<br>` WTKG3 = round((WTKG3*100),1);`<br>` IF _BMI5 NE . THEN _BMI5 = ROUND((_BMI5*100),1);` |

| Section 8: Demographics | | |
|---|---|---|
| **_CHLDCNT** | ***Calculated variable for number of children in household. _CHLDCNT is derived from CHILDREN.*** | |
| 1 | No children in household | Respondents who reported having no children. (CHILDREN=88) |
| 2 | One child in household | Respondents who reported having one child. (CHILDREN=1) |
| 3 | Two children in household | Respondents who reported having two children. (CHILDREN=2) |
| 4 | Three children in household | Respondents who reported having three children. (CHILDREN=3) |
| 5 | Four children in household | Respondents who reported having four children. (CHILDREN=4) |
| 6 | Five or more children in household | Respondents who reported having five or more children. (5 </= CHILDREN < 87) |
| 9 | Don't know/ Not sure/ Missing | Respondents who reported they didn´t know, were not sure, refused or had a missing value for CHILDREN. (CHILDREN=99) |
| | **SAS Code:** | `IF CHILDREN = 88 THEN _CHLDCNT = 1;`<br>` ELSE IF CHILDREN = 01 THEN _CHLDCNT = 2;`<br>` ELSE IF CHILDREN = 02 THEN _CHLDCNT = 3;`<br>` ELSE IF CHILDREN = 03 THEN _CHLDCNT = 4;`<br>` ELSE IF CHILDREN = 04 THEN _CHLDCNT = 5;`<br>`ELSE IF 05 <= CHILDREN < 88 THEN _CHLDCNT = 6;`<br>` ELSE IF CHILDREN = 99 THEN _CHLDCNT = 9;`<br>` ELSE IF CHILDREN = . THEN _CHLDCNT = 9;` |

Change to better reflect count for readability?

| Section 8: Demographics | | |
|---|---|---|
| **_EDUCAG** | ***Calculated variable for level of education completed.* _EDUCAG is derived from EDUCA.** | |
| 1 | Did not graduate High School | Respondents who reported they did not graduate high school. (EDUCA=1,2,3) |
| 2 | Graduated High School | Respondents who reported they graduated high school. (EDUCA=4) |
| 3 | Attended College or Technical School | Respondents who reported they attended college or technical school. (EDUCA=5) |
| 4 | Graduated from College or Technical School | Respondents who reported they graduated from college or technical school. (EDUCA=6) |
| 9 | Don't know/ Not sure/ Missing | Respondents who reported they didn´t know, were not sure, refused, or had a missing value for EDUCA. (EDUCA=9, missing) |
| | **SAS Code:** | `IF EDUCA IN (1,2,3) THEN _EDUCAG = 1;`<br>`ELSE IF EDUCA IN (4) THEN _EDUCAG = 2;`<br>`ELSE IF EDUCA IN (5) THEN _EDUCAG = 3;`<br>`ELSE IF EDUCA IN (6) THEN _EDUCAG = 4;`<br>`ELSE IF EDUCA IN (.,9) THEN _EDUCAG = 9;` |

| Section 8: Demographics | | |
|---|---|---|
| **_INCOMG1** | ***Calculated variable for income categories.* _INCOMG1 is derived from INCOME3.** | |
| 1 | Less than $15,000 | Respondents whose reported income is < $15,000. (INCOME3=1,2) |
| 2 | $15,000 to < $25,000 | Respondents whose reported income is $15,000 to < $25,000. (INCOME3=3,4) |
| 3 | $25,000 to < $35,000 | Respondents whose reported income is $25,000 to < $35,000. (INCOME3=5) |
| 4 | $35,000 to < $50,000 | Respondents whose reported income is $35,000 to < $50,000. (INCOME3=6) |
| 5 | $50,000 to < $100,000 | Respondents whose reported income is $50,000 to < $100,000. (INCOME3=7,8) |
| 6 | $100,000 to < $200,000 | Respondents whose reported income is $100,000 to < $200,000. (INCOME3=9,10) |
| 7 | $200,000 or more | Respondents whose reported income is $200,000 or more. (INCOME3=11) |
| 9 | Don't know/ Not sure/ Missing | Respondents who refused to answer, didn´t know, or had a missing value for INCOME3. (INCOME3=77, 99, or missing) |
| | **SAS Code:** | `IF INCOME3 IN (1,2) THEN _INCOMG1 = 1;`<br>`ELSE IF INCOME3 IN (3,4) THEN _INCOMG1 = 2;`<br>`ELSE IF INCOME3 IN (5) THEN _INCOMG1 = 3;`<br>`ELSE IF INCOME3 IN (6) THEN _INCOMG1 = 4;`<br>`ELSE IF INCOME3 IN (7,8) THEN _INCOMG1 = 5;`<br>`ELSE IF INCOME3 IN (9,10) THEN _INCOMG1 = 6;`<br>`ELSE IF INCOME3 IN (11) THEN _INCOMG1 = 7;`<br>`ELSE IF INCOME3 IN (77,99,.) THEN _INCOMG1 = 9;`<br>`ELSE IF INCOME3 IN (11)  THEN _INCOMG1 = 7;`<br>`ELSE IF INCOME3 IN (77,99,.) THEN _INCOMG1 = 9;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_MRACE1** | ***Calculated variable for calculated multiracial race categorization.*** **_MRACE1 is derived from MRACASC1. If respondents reported more than one race, they are assigned to the multiracial category. If MRACASC1 is less than 40 or equal to 60, then _MRACE1=MRACASC1. If MRACASC1 is 40-47 then _MRACE1=40. If MRACASC1 is 50-54 then _MRACE1=50.** | |
| 1 | White only | Respondents who reported they are white. (MRACASC1=10) |
| 2 | Black or African American only | Respondents who reported they are black. (MRACASC1=20) |
| 3 | American Indian or Alaska Native only | Respondents who reported they are American Indian or Alaska Native. (MRACASC1=30) |
| 4 | Asian Only | Respondents who reported they are Asian. (MRACASC1=40,41,42,43,44,45,46,47) |
| 5 | Native Hawaiian or other Pacific Islander only | Respondents who reported they are native Hawaiian or Pacific Islander. (MRACASC1=50,51,52,53,54) |
| 6 | Other race only | Respondents who reported they are of some other race group not listed in the question responses. (MRACASC1=60) |
| 7 | Multiracial | Respondents who reported they are of more than one race group (MRACASC1 > 99) |
| 77 | Don’t know/ Not sure | Respondents who reported they did not know their race. (MRACASC1=77) |
| 99 | Refused | Respondents who refused to give their race information. (MRACASC1=99) |
| | **SAS Code:** | `IF MRACASC1 GT 99 THEN _MRACE1 = 7;`<br>`ELSE IF MRACASC1 EQ 99 THEN _MRACE1 = 99;`<br>`ELSE IF MRACASC1 EQ 77 THEN _MRACE1 = 77;`<br>`ELSE IF MRACASC1 EQ 10 THEN _MRACE1 = 1;`<br>`ELSE IF MRACASC1 EQ 20 THEN _MRACE1 = 2;`<br>`ELSE IF MRACASC1 EQ 30 THEN _MRACE1 = 3;`<br>`ELSE IF 40 LE MRACASC1 LE 47 THEN _MRACE1 = 4;`<br>`ELSE IF 50 LE MRACASC1 LE 54 THEN _MRACE1 = 5;`<br>`ELSE IF MRACASC1=60 THEN _MRACE1=6;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_HISPANC** | ***Calculated variable for Hispanic, Latino/a, or Spanish origin calculated variable.*** **_HISPANC is derived from HISPANC3** | |
| 1 | Hispanic, Latino/a, or Spanish origin | Respondents who reported being of Hispanic, Latino/a, or Spanish origin (HISPANC3=1,2,3,4 or HISPANC3 > 9) |
| 2 | Not of Hispanic, Latino/a, or Spanish origin | Respondents who reported they were not of Hispanic, Latino/a, or Spanish origin (HISPANC3=5) |
| 9 | Don´t Know, Refused or Missing | Respondents who refused to report if they were of Hispanic, Latino/a, or Spanish origin (HISPANC3=7) |
| . | Not asked or Missing | Respondents who reported they did not know if they were of Hispanic, Latino/a, or Spanish origin (HISPANC3=9) |
| | **SAS Code:** | `HISPNUM=INPUT(HISPANC3,4.0);`<br>`IF HISPNUM in (5,58) THEN _HISPANC=2;`<br>`ELSE IF HISPNUM in (7,9,.) THEN _HISPANC=9;`<br>`ELSE _HISPANC=1;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_RACE** | ***Calculated variable for race/ethnicity categories.* _RACE is derived from _MRACE1 and _HISPANC. All respondents who reported they are of Hispanic or Latino origin are coded as Hispanic.** | |
| 1 | White only, non-Hispanic | Respondents who reported they are white and not of Hispanic origin. (_MRACE1=1 and _HISPANC=2) |
| 2 | Black only, non-Hispanic | Respondents who reported they are black and not of Hispanic origin. (_MRACE1=2 and _HISPANC=2) |
| 3 | American Indian or Alaska Native only, Non-Hispanic | Respondents who reported they are American Indian or Alaska Native and not of Hispanic origin. (_MRACE1=3 and _HISPANC=2) |
| 4 | Asian only, non-Hispanic | Respondents who reported they are Asian and not of Hispanic origin. (_MRACE1=4 and _HISPANC=2) |
| 5 | Native Hawaiian or other Pacific Islander only, Non-Hispanic | Respondents who reported they are Native Hawaiian or Pacific Islander and not of Hispanic origin. (_MRACE1=5 and _HISPANC=2) |
| 6 | Other race only, non-Hispanic | Respondents who reported they are of some other race group not listed in the question responses and are not of Hispanic origin. (_MRACE1=6 and _HISPANC=2) |
| 7 | Multiracial, non-Hispanic | Respondents who reported they are of more than one race group and are not of Hispanic origin. (_MRACE1=7 and _HISPANC=2) |
| 8 | Hispanic | Respondents who reported they are of Hispanic origin. ( _HISPANC=1) |
| 9 | Don’t know/ Not sure/ Refused | Respondents who reported they did not know or refused to give their race and are not of Hispanic origin or did not know or refused to answer if they are of Hispanic origin. (_MRACE1 =77, 99 and _HISPANC=2 or _HISPANC=7, 9) |
| | **SAS Code:** | `IF _HISPANC=9 OR (_MRACE1 IN(77,99) AND HISPANC3 EQ 2) THEN DO;`<br>`_RACE = 9 ;`<br>`END;`<br>`ELSE IF _HISPANC =2 THEN DO;`<br>`IF _MRACE1 = 1 THEN _RACE = 1 ;`<br>`ELSE IF _MRACE1 = 2 THEN _RACE = 2 ;`<br>`ELSE IF _MRACE1 = 3 THEN _RACE = 3 ;`<br>`ELSE IF _MRACE1 = 4 THEN _RACE = 4 ;`<br>`ELSE IF _MRACE1 = 5 THEN _RACE = 5 ;`<br>`ELSE IF _MRACE1 = 6 THEN _RACE = 6 ;`<br>`ELSE IF _MRACE1 = 7 THEN _RACE = 7 ;`<br>`END;`<br>`ELSE IF _HISPANC=1 THEN DO;`<br>`_RACE = 8 ;`<br>`END;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_RACEG21** | ***Calculated variable for white non-Hispanic race group.*** **_RACEG21 is derived from _RACE.** | |
| 1 | Non-Hispanic White | Respondents who reported they are white and not of Hispanic origin. (_RACE=1) |
| 2 | Non-White or Hispanic | Respondents who reported they are non-white or of Hispanic origin. (_RACE=2, 3, 4, 5, 6, 7, 8) |
| 9 | Don't know/ Not sure/ Refused | Respondents who reported they did not know or refused to give their race and are not of Hispanic origin or did not know or refused to answer if they are of Hispanic origin. (_RACE=9) |
| | **SAS Code:** | `IF _RACE = 1 THEN _RACEG21 = 1;`<br>` ELSE IF _RACE IN (2,3,4,5,6,7,8) THEN _RACEG21 = 2;`<br>` ELSE IF _RACE=9 THEN _RACEG21 = 9;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_RACEGR3** | ***Calculated variable for five-level race/ethnicity category.*** **_RACEGR3 is derived from _RACE.** | |
| 1 | White only, Non-Hispanic | Respondents who reported they are white and not of Hispanic origin. (_RACE=1) |
| 2 | Black only, Non-Hispanic | Respondents who reported they are black and not of Hispanic origin. (_RACE=2) |
| 3 | Other race only, Non-Hispanic | Respondents who reported they are not white and not black and not of Hispanic origin. (_RACE=3, 4, 5, 6) |
| 4 | Multiracial, Non-Hispanic | Respondents who reported being multiracial but not of Hispanic origin. (_RACE=7) |
| 5 | Hispanic | Respondents who reported they are of Hispanic origin. (_RACE=8) |
| 9 | Don't know/ Not sure/ Refused | Respondents who reported they did not know or refused to give their race and are not of Hispanic origin or did not know or refused to answer if they are of Hispanic origin. (_RACE=9) |
| | **SAS Code:** | `IF _RACE=1 THEN _RACEGR3=1;`<br>` ELSE IF _RACE=2 THEN _RACEGR3=2;`<br>` ELSE IF 3 LE _RACE LE 6 THEN _RACEGR3=3;`<br>` ELSE IF _RACE=7 THEN _RACEGR3=4;`<br>` ELSE IF _RACE=8 THEN _RACEGR3=5;`<br>` ELSE IF _RACE=9 THEN _RACEGR3=9;` |

| **Section 8: Demographics** | | |
|---|---|---|
| **_RACEPRV** | ***Calculated variable for computed race groups used for internet prevalence tables.*** **_RACEPRV is derived from _RACE and _IMPRACE** | |
| 1 | White only, non-Hispanic | Respondents who reported they are white and not of Hispanic origin or were imputed to be white and not of Hispanic origin. (_RACE=1 or _RACE=9 and _IMPRACE=1) |
| 2 | Black only, non-Hispanic | Respondents who reported they are black and not of Hispanic origin or were imputed to be black and not of Hispanic origin. (_RACE=2 or _RACE=9 and _IMPRACE=2) |
| 3 | American Indian or Alaska Native only, Non-Hispanic | Respondents who reported they are American Indian or Alaska Native and not of Hispanic origin or were imputed to be American Indian or Alaska Native and not of Hispanic origin. (_RACE=3 or _RACE=9 and _IMPRACE=4) |
| 4 | Asian only, non-Hispanic | Respondents who reported they are Asian and not of Hispanic origin or were imputed to be Asian and not of Hispanic origin. (_RACE=4 or _RACE=9 and _IMPRACE=3) |
| 5 | Native Hawaiian or other Pacific Islander only, Non-Hispanic | Respondents who reported they are Native Hawaiian or Pacific Islander and not of Hispanic origin. (_RACE=5) |
| 6 | Other race only, non-Hispanic | Respondents who reported they are of some other race group not listed in the question responses and are not of Hispanic origin or were imputed to be some other race group and not of Hispanic origin. (_RACE=6 or _RACE=9 and _IMPRACE=6) |
| 7 | Multiracial, non-Hispanic | Respondents who reported they are of more than one race group and are not of Hispanic origin. (_RACE=7) |
| 8 | Hispanic | Respondents who reported they are of Hispanic origin or were imputed to be of Hispanic origin. (_RACE=8 or _RACE=9 and _IMPRACE==5) |
| | **SAS Code:** | `IF _RACE < 9 THEN _RACEPRV=_RACE;`<br>`IF _RACE=9 THEN DO;`<br>`IF _IMPRACE IN (1,2,6) THEN _RACEPRV=_IMPRACE;`<br>`ELSE IF _IMPRACE=3 THEN _RACEPRV=4;`<br>`ELSE IF _IMPRACE=4 THEN _RACEPRV=3;`<br>`ELSE IF _IMPRACE=5 THEN _RACEPRV=8;`<br>`END;` |

**Section 9: Disability**

There are no calculated variables for Section 9.

**Section 10: Inadequate Sleep**

There are no calculated variables for Section 10.

| Section 11: Tobacco Use | | |
|---|---|---|
| **_SMOKER3** | ***Calculated variable for four-level smoker status: everyday smoker, someday smoker, former smoker, non-smoker.* _SMOKER3 is derived from SMOKE100 and SMOKDAY2.** | |
| 1 | Current smoker - now smokes every day | Respondents who reported having smoked at least 100 cigarettes in their lifetime and now smoke every day. (SMOKE100=1 and SMOKDAY2=1) |
| 2 | Current smoker - now smokes some days | Respondents who reported having smoked at least 100 cigarettes in their lifetime and now smoke some days. (SMOKE100=1 and SMOKDAY2=2) |
| 3 | Former smoker | Respondents who reported having smoked at least 100 cigarettes in their lifetime and currently do not smoke. (SMOKE100=1 and SMOKDAY2=3) |
| 4 | Never smoked | Respondents who reported they had not smoked at least 100 cigarettes in their lifetime. (SMOKE100=2) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they didn't know if they had smoked 100 cigarettes in their lifetime, those who refused to answer if they had smoked 100 cigarettes in their lifetime, those who didn't know if they now smoked every day, some days or not at all, those who refused to answer if they now smoked every day, some days or not at all, or those with missing responses. (SMOKE100=7, 9, missing; or SMOKDAY2=7, 9, missing) |
| | **SAS Code:** | `IF SMOKE100=2 THEN _SMOKER3=4;`<br>` ELSE IF SMOKE100=1 THEN DO;`<br>` IF SMOKDAY2=1 THEN _SMOKER3=1;`<br>` ELSE IF SMOKDAY2=2 THEN _SMOKER3=2;`<br>` ELSE IF SMOKDAY2 = 3 THEN _SMOKER3=3;`<br>` ELSE _SMOKER3=9;`<br>` END;`<br>` ELSE _SMOKER3=9;` |

| Section 11: Tobacco Use | | |
|---|---|---|
| **_RFSMOK3** | ***Calculated variable for adults who are current smokers.* _RFSMOK3 is derived from _SMOKER3.** | |
| 1 | No | Respondents who reported they had not smoked at least 100 cigarettes in their lifetime, those who reported having smoked 100 cigarettes in their lifetime but do not currently smoke. (_SMOKER3=3, 4) |
| 2 | Yes | Respondents who reported having smoked at least 100 cigarettes in their lifetime and currently smoke. (_SMOKER3=1, 2) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they did not know if they had smoked 100 cigarettes in their lifetime, those who refused to answer if they had smoked 100 cigarettes in their lifetime, those who didn't know if they now smoked every day, some days or not at all, those who refused to answer if they now smoked every day, some days or not at all, or those with missing responses. (_SMOKER3=9) |
| | **SAS Code:** | `IF _SMOKER3 IN (1,2) THEN _RFSMOK3=2;`<br>` ELSE IF _SMOKER3 IN (3,4) THEN _RFSMOK3=1;`<br>` ELSE _RFSMOK3=9;` |

| **Section 11: Tobacco Use** | | |
|---|---|---|
| **_CURECI3** | ***Calculated variable for adults who are current e-cigarette users.* _CURECI3 is derived from ECIGNOW3.** | |
| 1 | Not currently using E-cigarettes | Respondents who reported they had not used E-cigarettes in their lifetime, those who reported having used E-cigarettes in their lifetime but do not currently use E-cigarettes. (ECIGNOW3=1, 4) |
| 2 | Current E-cigarette user | Respondents who reported having used E-cigarettes in their lifetime and currently use E-cigarettes. (ECIGNOW3=2,3) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported they did not know if they had used E-cigarettes in their lifetime, those who refused to answer if they had used E-cigarettes in their lifetime, those who didn't know if they now used E-cigarettes every day, some days or not at all, those who refused to answer if they now used E-cigarettes every day, some days or not at all, or those with missing responses. (ECIGNOW3=7,9, or missing) |
| | **SAS Code:** | `IF ECIGNOW3 IN (2,3) THEN _CURECI3=2;`<br>`ELSE IF ECIGNOW3 IN (1,4) THEN _CURECI3=1;`<br>`ELSE _CURECI3=9;` |

| **Section 12: Alcohol Consumption** | | |
|---|---|---|
| **DRNKANY6** | ***Calculated variable for adults who reported having had at least one drink of alcohol in the past 30 days.* DRNKANY6 is derived from ALCDAY4** | |
| 1 | Yes | Respondents who reported drinking at least one alcoholic beverage in the past 30 days. (1 </= ALCDAY4 </= 231) |
| 2 | No | Respondents who reported drinking no alcoholic beverages in the past 30 days. (ALCDAY4=888) |
| 7 | Don't know/ Not Sure | Respondents who reported not knowing if they drank at least one alcoholic beverage in the past 30 days. (ALCDAY4=777) |
| 9 | Refused/ Missing | Respondents who refused to answer or had a missing value for drinking at least one alcoholic beverage in the past 30 days. (ALCDAY4=999, Missing) |
| | **SAS Code:** | `IF 1 <= ALCDAY4 < 231 THEN DRNKANY6=1;`<br>`ELSE IF ALCDAY4=888 THEN DRNKANY6=2;`<br>`ELSE IF ALCDAY4=777 THEN DRNKANY6=7;`<br>`ELSE DRNKANY6=9;` |

| Section 12: Alcohol Consumption | | |
|---|---|---|
| **DROCDY4_** | ***Calculated variable for drink-occasions-per-day.* DROCDY4_ is derived from ALCDAY4 by dividing the ALCDAY4 variable by 7 days per week or 30 days per month.** | |
| 0 | No Drink-Occasions per day | Respondents reported no occasions per day that they consumed alcohol. (ALCDAY4=888) |
| 1 - 899 | Drink-Occasions per day | Respondents reported number of occasions per day that they consumed alcohol. (ALCDAY4 not equal to 777, 888, 999, or missing) |
| 900 | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they did not know how many days they had at least one drink of alcohol, those who refused to answer how many days they had at least one drink of alcohol, those with missing responses. (ALCDAY4=777, 999, or missing) |
| | **SAS Code:** | `IF ALCDAY4 NOTIN (888,777,999,.) THEN DO;`<br>` IF 101 LE ALCDAY4 LE 107 THEN DROCDY4_=(ALCDAY4-100)/7;`<br>` ELSE IF 201 LE ALCDAY4 LE 230 THEN DROCDY4_=(ALCDAY4-200)/30;`<br>` END;`<br>`ELSE IF ALCDAY4 EQ 888 THEN DROCDY4_=0;`<br>` ELSE DROCDY4_=9;`<br>`* DROCDY4_=round((DROCDY4_*100),1);`<br>` *This is done after all of the alcohol calculations but the code is included here;` |

| Section 12: Alcohol Consumption | | |
|---|---|---|
| **_RFBING6** | ***Calculated variable for binge drinkers (males having five or more drinks on one occasion, females having four or more drinks on one occasion).* _RFBING6 is derived from DRNK3GE5 and ALCDAY4.** | |
| 1 | No | Respondents who reported they did not drink in the past 30 days, or those who reported that they did drink alcohol in the past 30 days but did not report having five or more drinks of alcohol on an occasion. (ALCDAY4 < 231 and DRNK3GE5=88; or ALCDAY4=888) |
| 2 | Yes | Respondents who reported they did drink in the past 30 days and had five or more drinks on one or more occasions in the past month. (ALCDAY4 < 231 and 1 </= DRNK3GE5 </= 76) |
| 9 | Don't know/ Refused/ Missing | Respondents who reported that they did not know if they had consumed five or more drinks of alcohol on one occasion or refused to answer if they had consumed five or more drinks of alcohol on one occasion or those with missing responses. (DRNK3GE5=77, 99, missing; or ALCDAY4=777, 999, missing) |
| | **SAS Code:** | `IF ALCDAY4 NOTIN (888) THEN DO;`<br>` IF 1 LE DRNK3GE5 LE 76 THEN _RFBING6=2;`<br>` ELSE IF DRNK3GE5 IN (.,77,99) THEN _RFBING6=9;`<br>` ELSE IF DRNK3GE5 IN (88) THEN _RFBING6=1;`<br>` END;`<br>`ELSE IF ALCDAY4 = 888 THEN _RFBING6=1;`<br>`  ELSE _RFBING6=9;` |

| Section 12: Alcohol Consumption | | |
|---|---|---|
| **_DRNKWK3** | ***Calculated variable for calculated total number of alcoholic beverages consumed per week.*** **_DRNKWK3 is derived from DROCDY4_ and AVEDRNK4 by multiplying the total number of drink occasions per day (DROCDY4_) by the average number of drinks per occasion (AVEDRNK4) times seven days.** | |
| 0 | Did not drink | Respondents who did not drink in the past month. (DROCDY4_=0 or AVEDRNK4=88) |
| 1 - 98999 | Number of drinks per week | Respondents reported number of alcoholic drinks in the past week. (0 < DROCDY4_ < 990) |
| 99900 | Don't know/ Not sure/ Refused/ Missing | Respondents who refused to report the number of alcohol drinks consumed per day, or respondents who did not know the number of alcohol drinks consumed per day, or those with missing responses or respondents who refused to report the number drink occasions per day, or respondents who did not know the number of drink occasions per day, or those with missing responses. (AVEDRNK4=.,77,99 or DROCDY4_=900) |
| | **SAS Code:** | `IF DROCDY4_=0 THEN _DRNKWK3=0;`<br>`ELSE IF DROCDY4_=9 THEN _DRNKWK3=999;`<br>`ELSE IF AVEDRNK4 IN (.,77,99) THEN _DRNKWK3=999;`<br>`ELSE IF AVEDRNK4=88 THEN _DRNKWK3=0;`<br>`ELSE _DRNKWK3=AVEDRNK4*DROCDY4_*7;`<br>`* _DRNKWK3=ROUND((_DRNKWK3*100),1);`<br>`*This is done after all of the alcohol calculations but the code is included here;` |

| **Section 12: Alcohol Consumption** | | |
|---|---|---|
| **_RFDRHV9** | ***Calculated variable for heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week).* _RFDRHV9 is derived from _DRNKWK3, ALCDAY4, and SEXVAR.** | |
| 1 | No | Male Respondents who reported having 14 drinks per week or less, or Female Respondents who reported having 7 drinks per week or less. ((SEXVAR=1 or BIRTHSEX=1) and _DRNKWK3 </= 1400 or (SEXVAR=2 or BIRTHSEX=2) and _DRNKWK3 </= 700 or ALCDAY4=888) |
| 2 | Yes | Male Respondents who reported having more than 14 drinks per week, or Female Respondents who reported having more than 7 drinks per week. ((SEXVAR=1 or BIRTHSEX=1) and _DRNKWK3 > 1400 or (SEXVAR=2 or BIRTHSEX=2) and _DRNKWK3 > 700) |
| 9 | Don't know/ Refused/ Missing | Respondents with don´t know, refused or missing responses for ALCDAY4 or _DRNKWK3. (ALCDAY5=777, 999, or missing, or _DRNKWK3=99, or missing) |
| | **SAS Code:** | `IF (SEXVAR=1 or BIRTHSEX=1) AND _DRNKWK3 NOTIN (999,.) THEN DO;`<br>`IF _DRNKWK3 GT 14 THEN _RFDRHV9=2;`<br>`ELSE IF _DRNKWK3 LE 14 THEN _RFDRHV9=1;`<br>`END;`<br>`ELSE IF (SEXVAR=2 or BIRTHSEX=2) AND _DRNKWK3 NOTIN (999,.) THEN DO;`<br>`IF _DRNKWK3 GT 7 THEN _RFDRHV9=2;`<br>`ELSE IF _DRNKWK3 LE 7 THEN _RFDRHV9=1;`<br>`END;`<br>`ELSE IF ALCDAY4 EQ 888 THEN _RFDRHV9=1;`<br>`ELSE _RFDRHV9=9;`<br>`** ROUND OFF TO NO DECIMAL PLACES ** MULTIPLY BY 100 AND THEN ROUND OFF TO NO DECIMAL PLACES AND THEN REMOVE THE DECIMAL PLACES **;`<br>`DROCDY4_=round((DROCDY4_*100),1);`<br>`_DRNKWK3=ROUND((_DRNKWK3*100),1);` |

| **Section 13: Immunization** | | |
|---|---|---|
| **_FLSHOT7** | ***Calculated variable for adults aged 65+ who have had a flu shot within the past year.*** **_FLSHOT7 is derived from FLUSHOT7.** | |
| 1 | Yes | Respondents aged 65 or older who reported having a flu shot within the past 12 months. (AGE >/= 65 and FLUSHOT7=1) |
| 2 | No | Respondents aged 65 or older who reported not having had a flu shot within the past 12 months. (AGE >/= 65 and FLUSHOT7=2) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who did not know their age, those who refused to report their age, those who didn't know if they had a flu shot in the past 12 months, or those who refused to answer if they had a flu shot in the past 12 months, or those with missing responses. (AGE >/= 65 and FLUSHOT7=7,9, or missing or AGE=7,9, or missing) |
| . | Age < 65 | Respondents aged 18-64. (18 </= AGE </= 64) |
| | **SAS Code:** | `IF AGE GE 65 THEN DO;`<br>` IF FLUSHOT7=1 THEN _FLSHOT7=1;`<br>` ELSE IF FLUSHOT7=2 THEN _FLSHOT7=2;`<br>` ELSE IF FLUSHOT7 IN (.,7,9) THEN _FLSHOT7=9;`<br>` END;`<br>` ELSE IF AGE IN (.,7,9) THEN _FLSHOT7=9;`<br>` ELSE _FLSHOT7=.;` |

| **Section 13: Immunization** | | |
|---|---|---|
| **_PNEUMO3** | ***Calculated variable for adults aged 65+ who have ever had a pneumonia vaccination.*** **_PNEUMO3 is derived from PNEUVAC4.** | |
| 1 | Yes | Respondents aged 65 or older who reported having a pneumonia shot. (AGE >/= 65 and PNEUVAC4=1) |
| 2 | No | Respondents aged 65 or older who reported not having had a pneumonia shot. (AGE >/= 65 and PNEUVAC4=2) |
| 9 | Don't know/ Not Sure Or Refused/ Missing | Respondents who did not know their age, those who refused to report their age, those who did not know if they ever had a pneumonia shot, those who refused to answer if they had a pneumonia shot, or those with missing responses. (AGE >/= 65 and PNEUVAC3=7,9, or missing or AGE=7,9, or missing) |
| . | Age < 65 | Respondents aged 18-64. (18 </= AGE </= 64) |
| | **SAS Code:** | `IF AGE GE 65 THEN DO;`<br>` IF PNEUVAC4=1 THEN _PNEUMO3=1;`<br>` ELSE IF PNEUVAC4=2 THEN _PNEUMO3=2;`<br>` ELSE IF PNEUVAC4 IN (.,7,9) THEN _PNEUMO3=9;`<br>` ELSE _PNEUMO3=.;`<br>` END;`<br>` ELSE IF AGE IN (.,7,9) THEN _PNEUMO3=9;`<br>` ELSE _PNEUMO3=.;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **FTJUDA2_** | ***Calculated variable for fruit juice intake in times per day.*** **FTJUDA2_ converts the FRUITJU2 variable to a per day response. (Two implied decimal places)** | |
| 0 - 9999 | Times per day (two implied decimal places) | Respondents reported intake of fruit juice per day (FRUITJU2 not equal to 777,999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know the number of times fruit juice was consumed per day, those who refused to answer, and those with missing responses (FRUITJU2=777,999, or missing) |
| | **SAS Code:** | `IF 100 < FRUITJU2 < 200 THEN FTJUDA2_=FRUITJU2-100;`<br>`ELSE IF 200 < FRUITJU2 < 300 THEN FTJUDA2_=(ROUND((FRUITJU2-200)/7,0.01));`<br>`ELSE IF 300 < FRUITJU2 < 400 THEN FTJUDA2_=(ROUND((FRUITJU2-300)/30,0.01));`<br>`ELSE IF FRUITJU2 = 555 THEN FTJUDA2_=0;`<br>`ELSE IF FRUITJU2 = 300 THEN FTJUDA2_=0.02;`<br>`ELSE IF FRUITJU2 IN (.,777,999) THEN FTJUDA2_=.;`<br>`** ROUND OFF **;`<br>`FTJUDA2_=round((FTJUDA2_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **FRUTDA2_** | ***Calculated variable for fruit intake in times per day.*** **FRUTDA2_ converts the FRUIT2 variable to a per day response. (Two implied decimal places)** | |
| 0 - 9999 | Times per day (two implied decimal places) | Respondents reported intake of fruit per day (FRUIT2 not equal to 777,999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know the number of times fruit was consumed per day, those who refused to answer, and those with missing responses (FRUIT2=777, 999, or missing) |
| | **SAS Code:** | `IF 100 < FRUIT2 < 200  THEN FRUTDA2_=FRUIT2-100;`<br>`ELSE IF 200 < FRUIT2 < 300  THEN FRUTDA2_=(ROUND((FRUIT2-200)/7,0.01));`<br>`ELSE IF 300 < FRUIT2 < 400  THEN FRUTDA2_=(ROUND((FRUIT2-300)/30,0.01));`<br>`ELSE IF FRUIT2 = 555 THEN FRUTDA2_=0;`<br>`ELSE IF FRUIT2 = 300 THEN FRUTDA2_=0.02;`<br>`ELSE IF FRUIT2 IN (.,777,999) THEN FRUTDA2_=.;`<br>`** ROUND OFF **;`<br>`FRUTDA2_=round((FRUTDA2_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **GRENDA1_** | ***Calculated variable for dark green vegetable intake in times per day.*** **GRENDA1_ converts the FVGREEN1 variable to a per day response (Two implied decimal places)** | |
| 0 - 9999 | Times per day (two implied decimal places) | Respondents reported intake of dark green vegetables per day (FVGREEN1 not equal to 777,999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know the number of times dark green vegetables were consumed per day, those who refused to answer, and those with missing responses (FVGREEN1=777,999, or missing) |
| | **SAS Code:** | `IF 100 < FVGREEN1 < 200 THEN GRENDA1_=FVGREEN1-100;`<br>`ELSE IF 200 < FVGREEN1 < 300 THEN GRENDA1_=(ROUND((FVGREEN1-200)/7,0.01));`<br>`ELSE IF 300 < FVGREEN1 < 400 THEN GRENDA1_=(ROUND((FVGREEN1-300)/30,0.01));`<br>`ELSE IF FVGREEN1 = 555 THEN GRENDA1_=0;`<br>`ELSE IF FVGREEN1 = 300 THEN GRENDA1_=0.02;`<br>`ELSE IF FVGREEN1 IN (.,777,999) THEN GRENDA1_=.;`<br>`** ROUND OFF **;`<br>`GRENDA1_=round((GRENDA1_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **FRNCHDA_** | ***Calculated variable for French fry intake in times per day.*** **FRNCHDA_ converts the FRENCHF1 variable to a per day response. (Two implied decimal places)** | |
| 0 - 9999 | Times per day (two implied decimal places) | Respondents reported intake of French fries per day (FRENCHF1 not equal to 777,999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn´t know the number of times French fries were consumed per day, those who refused to answer, and those with missing responses (FRENCHF1=777, 999, or missing) |
| | **SAS Code:** | `IF 100 < FRENCHF1 < 200 THEN FRNCHDA_=FRENCHF1-100;`<br>`ELSE IF 200 < FRENCHF1 < 300 THEN FRNCHDA_=(ROUND((FRENCHF1-200)/7,0.01));`<br>`ELSE IF 300 < FRENCHF1 < 400 THEN FRNCHDA_=(ROUND((FRENCHF1-300)/30,0.01));`<br>`ELSE IF FRENCHF1 = 555 THEN FRNCHDA_=0;`<br>`ELSE IF FRENCHF1 = 300 THEN FRNCHDA_=0.02;`<br>`ELSE IF FRENCHF1 IN (.,777,999) THEN FRNCHDA_=.;`<br>`** ROUND OFF **;`<br>`FRNCHDA_=round((FRNCHDA_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **POTADA1_** | ***Calculated variable for potato servings per day.*** **POTADA1_ converts the POTATOE1 variable to a per-day response.** | |
| 0 - 9999 | Times per day | Respondents reported servings of potatoes per day (POTATOE1 not equal to 777, 999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn't know the quantity of potato servings consumed per day, those who refused to answer, and those with missing responses (POTATOE1=777, 999, or missing) |
| | **SAS Code:** | `IF 100 < POTATOE1 < 200 THEN POTADA1_=POTATOE1-100;`<br>`ELSE IF 200 < POTATOE1 < 300 THEN POTADA1_=(ROUND((POTATOE1-200)/7,0.01));`<br>`ELSE IF 300 < POTATOE1 < 400 THEN POTADA1_=(ROUND((POTATOE1-300)/30,0.01));`<br>`ELSE IF POTATOE1 = 555 THEN POTADA1_=0;`<br>`ELSE IF POTATOE1 = 300 THEN POTADA1_=0.02;`<br>`ELSE IF POTATOE1 IN (.,777,999) THEN POTADA1_=.;`<br>`** ROUND OFF **;`<br>`POTADA1_=round((POTADA1_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **VEGEDA2_** | ***Calculated variable for other vegetable intake in times per day.*** **VEGEDA2_ converts the VEGETAB2 variable to a per day response. (Two implied decimal places)** | |
| 0 - 9999 | Times per day (two implied decimal places) | Respondents reported intake of other vegetables per day (VEGETAB2 not equal to 777, 999, or missing) |
| . | Don't know/ Not Sure Or Refused/ Missing | Respondents who reported they didn't know the number of times other vegetables were consumed per day, those who refused to answer, and those with missing responses (VEGETAB2=777, 999, or missing) |
| | **SAS Code:** | `IF 100 < VEGETAB2 < 200 THEN VEGEDA2_=VEGETAB2-100;`<br>`ELSE IF 200 < VEGETAB2 < 300 THEN VEGEDA2_=(ROUND((VEGETAB2-200)/7,0.01));`<br>`ELSE IF 300 < VEGETAB2 < 400 THEN VEGEDA2_=(ROUND((VEGETAB2-300)/30,0.01));`<br>`ELSE IF VEGETAB2 = 555 THEN VEGEDA2_=0;`<br>`ELSE IF VEGETAB2 = 300 THEN VEGEDA2_=0.02;`<br>`ELSE IF VEGETAB2 IN (.,777,999) THEN VEGEDA2_=.;`<br>`** ROUND OFF **;`<br>`VEGEDA2_=round((VEGEDA2_*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_MISFRT1** | ***Calculated variable for the number of missing fruit responses.*** **_MISFRT1 is derived from MFTJUDA2_ and MFRUTDA2_** | |
| 0 | No missing fruit responses | Respondents with no missing fruit responses |
| 1 - 2 | Has 1 or 2 missing fruit responses | Respondents with missing fruit responses |
| | **SAS Code:** | `IF FTJUDA2_=. THEN MFTJUDA2_=1;`<br>`ELSE MFTJUDA2_=0;`<br>`IF FRUTDA2_=. THEN MFRUTDA2_=1;`<br>`ELSE MFRUTDA2_=0;`<br>`_MISFRT1=SUM(MFTJUDA2_, MFRUTDA2_);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_MISVEG1** | ***Calculated variable for the number of missing vegetable responses.*** **_MISVEG1 is derived from MGRENDA1_, MFRNCHDA_, MPOTADA1_ and MVEGEDA2_.** | |
| 0 | No missing vegetable responses | Respondents with no missing vegetable responses |
| 1 - 4 | Has 1, 2, 3, or 4 missing vegetable responses | Respondents with missing vegetable responses |
| | **SAS Code:** | `IF GRENDA1_=. THEN MGRENDA1_=1;`<br>`ELSE MGRENDA1_=0;`<br>`IF FRNCHDA_=. THEN MFRNCHDA_=1;`<br>`ELSE MFRNCHDA_=0;`<br>`IF POTADA1_=. THEN MPOTADA1_=1;`<br>`ELSE MPOTADA1_=0;`<br>`IF VEGEDA2_=. THEN MVEGEDA2_=1;`<br>`ELSE MVEGEDA2_=0;`<br>`_MISVEG1=SUM(MGRENDA1_, MFRNCHDA_, MPOTADA1_, MVEGEDA2_);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_FRTRES1** | ***Calculated variable for missing any fruit responses.*** **_FRTRES1 is derived from _MISFRT1** | |
| 0 | Not Included - Missing Fruit Responses | Respondents with a missing value for one of the fruit variables (1 </= _MISFRT1 </= 2) |
| 1 | Included - Not Missing Fruit Responses | Respondents with no missing fruit variables (_MISFRT1=0) |
| | **SAS Code:** | `_FRTRES1=0;`<br>`IF 1<=_MISFRT1<=2 THEN _FRTRES1=0;`<br>`ELSE IF _MISFRT1=0 THEN _FRTRES1=1;` |

| **Section 14: Fruits and Vegetables** | | |
|---|---|---|
| **_VEGRES1** | ***Calculated variable for missing any vegetable responses.* _VEGRES1 is derived from GRENDA1_ , FRNCHDA_ , POTADA1_ , VEGEDA1_ and _MISVEG1.** | |
| 0 | Not Included - Missing Vegetable Responses | Respondents with missing vegetable per day values (1 </= _MISVEG1 </= 4) |
| 1 | Included - Not Missing Vegetable Responses | Respondents with no missing vegetable per day values (_MISVEG1=0) |
| . | Not asked or Missing | Respondents with a 99 value for all vegetable per day variables. |
| | **SAS Code:** | `_VEGRES1=0;`<br>`IF 1<=_MISVEG1<=4 THEN _VEGRES1=0;`<br>`ELSE IF _MISVEG1=0 THEN _VEGRES1=1;` |

| **Section 14: Fruits and Vegetables** | | |
|---|---|---|
| **_FRUTSU1** | ***Calculated variable for total fruits consumed per day.* _FRUTSU1 is derived from the individual fruit variables (FTJUDA2_, FRUTDA2_). Values for don´t know, refused, or missing" (99) are excluded from the sum.** | |
| 0 - 99998 | Number of Fruits consumed per day (two implied decimal places) | Number of Fruits consumed per day (two implied decimal places) (FTJUDA2_+FRUTDA2_) |
| . | Not asked or Missing | Respondents with a 99 value for all four fruits per day variables. |
| | **SAS Code:** | `_FRUTSU1=(FTJUDA2_/100) + (FRUTDA2_/100);`<br>`_FRUTSU1=round((_FRUTSU1*100),1);` |

| **Section 14: Fruits and Vegetables** | | |
|---|---|---|
| **_VEGESU1** | ***Calculated variable for total vegetables consumed per day.* _VEGESU1 is derived from the individual vegetable variables (GRENDA1_, FRNCHDA_, POTADA1_, and VEGEDA2_). Values for don´t know, refused, or missing" (99) are excluded from the sum.** | |
| 0 - 99998 | Number of Vegetables consumed per day (two implied decimal places) | Sum of all vegetable per day values (two implied decimal places) (GRENDA1_+FRNCHDA_+POTADA1_+VEGEDA2_) |
| . | Not asked or Missing | Respondents with a 99 value for all vegetable per day variables. |
| | **SAS Code:** | `_VEGESU1=(GRENDA1_/100) + (FRNCHDA_/100) + (POTADA1_/100) + (VEGEDA2_/100);`<br>`_VEGESU1=round((_VEGESU1*100),1);` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_FRTLT1A** | ***Calculated variable for consumed fruit 1 or more times per day.*** **_FRTLT1A is derived from _FRUTSU1** | |
| 1 | Consumed fruit one or more times per day | Respondents who reported consuming fruit 1 or more times a day (_FRUTSU1/100 >/= 1) |
| 2 | Consumed fruit < one time per day | Respondents who reported consuming fruit < 1 time a day (_FRUTSU1/100 < 1) |
| 9 | Don´t know, refused or missing values | Respondents with don´t know, not sure, refused, or missing responses (_FRUTSU1=.) |
| | **SAS Code:** | `IF 0 <= (_FRUTSU1/100) < 1 THEN _FRTLT1A=2;`<br>`ELSE IF (_FRUTSU1/100) >= 1 THEN _FRTLT1A=1;`<br>`ELSE _FRTLT1A=9;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_VEGLT1A** | ***Calculated variable for consumed vegetables 1 or more times per day.*** **_VEGLT1A is derived from _VEGESU1** | |
| 1 | Consumed vegetables one or more times per day | Respondents who reported consuming vegetables 1 or more times a day (_VEGESU1/100 >/= 1) |
| 2 | Consumed vegetables < one time per day | Respondents who reported consuming vegetables < 1 time a day (_VEGESU1/100 < 1) |
| 9 | Don´t know, refused or missing values | Respondents with don´t know, not sure, refused, or missing responses (_VEGESU1=.) |
| | **SAS Code:** | `IF 0 <= (_VEGESU1/100) < 1 THEN _VEGLT1A=2;`<br>`ELSE IF (_VEGESU1/100) >= 1 THEN _VEGLT1A=1;`<br>`ELSE _VEGLT1A=9;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_FRT16A** | ***Calculated variable for reported consuming fruit >16 per day.*** **_FRT16A is derived from _FRUTSU1** | |
| 0 | Not Included - Values are too high | Respondents with an out-of-range value for sum of fruits per day (_FRUTSU1 > 16) |
| 1 | Included - Values are in accepted range | Respondents with value for sum of fruits per day in acceptable range (_FRUTSU1 </= 16) |
| . | Not asked or Missing | Respondents with a 99 value for both fruit per day variables. |
| | **SAS Code:** | `IF (_FRUTSU1/100)>16 THEN _FRT16A=0;`<br>`ELSE IF (_FRUTSU1/100)<=16 THEN _FRT16A=1;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_VEG23A** | ***Calculated variable for reported consuming vegetables >23 per day.*** **_VEG23A is derived from _VEGESU1** | |
| 0 | Not Included - Values are too high | Respondents with an out-of-range value for sum of vegetables per day (_VEGESU1 > 23) |
| 1 | Included - Values are in accepted range | Respondents with value for sum of vegetables per day in acceptable range (_VEGESU1 </= 23) |
| . | Not asked or Missing | Respondents with a 99 value for all vegetable per day variables. |
| | **SAS Code:** | `IF (_VEGESU1/100)>23 THEN _VEG23A=0;`<br>`ELSE IF (_VEGESU1/100)<=23 THEN _VEG23A=1;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_FRUITE1** | ***Calculated variable for fruit exclusion from analyses.*** **_FRUITE1 is derived from _FRTRES1 and _FRT16A** | |
| 0 | No missing values and in accepted range | Respondents with no missing fruit values and in accepted range (_FRTRES1=1 AND _FRT16A=1) |
| 1 | Missing Fruit responses | Respondents missing at least one fruit per day value (_FRTRES1=0) |
| 2 | Fruit values out of range | Respondents with an out-of-range value for sum of fruits per day (_FRTRES1=1 AND _FRT16A=0) |
| . | Not asked or Missing | Respondents with a 99 value for both fruit per day variables. |
| | **SAS Code:** | `IF _FRTRES1=1 AND _FRT16A=0 THEN _FRUITE1=2;`<br>`ELSE IF _FRTRES1=1 AND _FRT16A=1 THEN _FRUITE1=0;`<br>`ELSE _FRUITE1=1;` |

| Section 14: Fruits and Vegetables | | |
|---|---|---|
| **_VEGETE1** | ***Calculated variable for vegetable exclusion from analyses.*** **_VEGETE1 is derived from _VEGRES1 and _VEG23A.** | |
| 0 | No missing values and in accepted range | Respondents with no missing vegetable per day values and in all accepted range (_VEGRES1=1 AND _VEG23A=1) |
| 1 | Missing Vegetable responses | Respondents with missing vegetable per day values (_VEGRES1=0) |
| 2 | Vegetable values out of range | Respondents with out-of-range vegetable per day values (_VEGRES1=1 AND _VEG23A=0) |
| . | Not asked or Missing | Respondents with a 99 value for all vegetable per day variables. |
| | **SAS Code:** | `IF _VEGRES1=1 AND _VEG23A=0 THEN _VEGETE1=2;`<br>`ELSE IF _VEGRES1=1 AND _VEG23A=1 THEN _VEGETE1=0;`<br>`ELSE _VEGETE1=1;` |

| **Section 15: HIV/AIDS** | | |
|---|---|---|
| **_AIDTST4** | ***Calculated variable for adults who have ever been tested for HIV.*** **_AIDTST4 is derived from HIVTST7.** | |
| 1 | Yes | Respondents who reported to having been tested for HIV. (HIVTST7=1) |
| 2 | No | Respondents who did not report having been tested for HIV. (HIVTST7=2) |
| 9 | Don’t know/ Not Sure/ Refused | Respondents who reported they did not know if they had been tested for HIV, or those who refused to answer if they had been tested for HIV. (HIVTST7=7,9) |
| . | Not asked or missing | Respondents with missing responses for HIVTST7. (HIVTST7=missing) |
| | **SAS Code:** | `IF HIVTST7=1 THEN _AIDTST4=1;`<br>`ELSE IF HIVTST7=2 THEN _AIDTST4=2;`<br>`ELSE IF HIVTST7 IN (7,9) THEN _AIDTST4=9;`<br>`ELSE IF HIVTST7=. THEN _AIDTST4=.;` |